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Excess of Neurons in the Human Newborn Mediodorsal Thalamus Compared with That of the Adult

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The aim of this study was to quantify the total number of neurons and glial cells in the mediodorsal nucleus of the thalamus (MD) of 8 newborn human brains, in comparison to 8 adult human brains. The estimates of the cell numbers were obtained using the stereological principles of the optical fractionator. In the case of the adults, the total number of neurons in the entire MD was an average of 41% lower than in the newborn, which was statistically highly significant ($P < 0.001$). The estimated average total number of neurons in MD thalamus of the newborns was 11.2 million (coefficient of variation [CV] = standard deviation/mean = 0.16), compared with the adults' 6.43 million (CV = 0.15). The glial cell numbers were substantially higher in the adult brains, with an increase of almost 4 times from 10.6 million at birth to 36.3 million in the fully developed adult brain. This is the first demonstration of a higher number of human neurons in the brain of newborns compared with the adult.

Keywords: cell numbers, development, disector, fractionator, neuron loss, stereology

Introduction

The human brain is distinguished by marked development in both size and complexity of the cerebral cortex and by a concomitant elaboration of interactions between the cerebral cortex and a main source of it's input, the thalamus. The mediodorsal nucleus of the thalamus (MD) manifests the largest relative difference in numbers of neurons when comparing apes and humans. This suggests that the function of this nucleus is more important for a particular aspect of human behavior than it is for other anthropoids (Hirai and Jones 1989; Armstrong 1990). Monakov (1895) suggested the MD as being linked to the cognitive and emotional life of an individual. Since then it has become evident that MD is the principal relay nucleus for the prefrontal cortex, which plays a major role in the development of human personality.

The input selection and output tuning of the MD are well documented (Jones 1997c), providing numerous reciprocal connections to the dorsolateral prefrontal cortex and have afferent connections coming from the striatum, the basal ganglia, and the limbic system (Goldman-Rakic and Porrino 1985; Armstrong 1990; Jones 1997a, 2007c). These connections enable the MD to bridge important subcortical and cortical areas providing it with the capacity to integrate cognitive activity with affective experience.

Several studies concerning the anatomy, projections, and afferent input to the MD have been the basis of a topographical map of the nucleus (Dewulf 1971; Goldman-Rakic and Porrino 1985; Hirai and Jones 1989; Armstrong 1990; Jones 1997a). Three different subregions (or subnuclei) have been ascribed to the MD; a magnocellular (MDmc), a parvocellular (MDpc), and a densocellular (MDdc) or multiform portion. The nomenclature for these subregions varies throughout the literature, and even though it partly describes general morphological characteristics of a given region, it may not consistently apply to the neuronal size or packing density (Armstrong 1990). However, each region has its own specific projections and afferent input, which ratifies the segregation (Goldman-Rakic and Porrino 1985; Armstrong 1990; Jones 1997a). MDmc is not only dominated by input from the olfactory and entorhinal cortices but also receives input from the amygdala and projects to the ventromedial and orbital area in the frontal lobe. MDpc receives input from the superior colliculus and other midbrain structures and projects to the dorsal and lateral areas of the prefrontal cortex. MDdc, in which we include the paralamellar part of MD, has its inputs from many of the same subcortical sites as the central lateral nucleus (CL) and projects to the striatum and areas adjacent to premotor cortex including the frontal eye fields. Inhibitory input reaches all 3 subregions from the ventral pallidum, the pars reticulata of substantia nigra, and the thalamic reticular nucleus. The non-specific brainstem cholinergic and monoaminergic systems also distribute fibers to the MD as they do to all thalamic nuclei.

Several studies have described alterations in the total number of neurons in the MD of different diagnostic groups. Among the most compelling findings is a reduction of 27–40% in the total number of neurons in chronic schizophrenic subjects (Pakkenberg 1990, 1992; Popken et al. 2000; Young et al. 2000; Byne et al. 2002), whereas other postmortem studies have not found volume or neuron number reduction in the MD (Rosenthal and Bigelow 1972; Lesch and Bogerts 1984; Falke et al. 2000; Cullen et al. 2003; Dorph-Petersen et al. 2004; Danos et al. 2005). An elevated neuron number of as much as 37% in subjects with major depressive disorder has also been documented (Young et al. 2004). These differences may reflect the disturbance of a normal developmentally regulated pattern of neuronal death and survival.

Knowledge about the human newborn brain can serve as a normative reference in the analysis of developmental neuroanatomy and be the basis for further understanding of neurogenesis or neuronal cell loss during human life. Consequently, the aim of this study was to quantify the total numbers of neurons and glial cells in the human MD at the time of birth and compare that number to the total numbers in the same nucleus of the adult brain, using stereological methods.

Materials and Methods

Subjects

The material comprised 8 normal newborn human brains, 4 males and 4 females, with gestational ages ranging from 38 to 41 weeks. For

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comparison, 8 normal adult human brains were included, 3 males and 5 females. A complete brain autopsy was performed in each of the adult cases within 24–48 h postmortem, including histological diagnosis and a thorough neuropathological examination of one hemisphere. The other hemisphere was preserved for stereological analysis. From all adult brains, tissue for neuropathological examination included the 4 cortical lobes, the insula, the gyrus cinguli, and the hippocampus. Immunohistochemistry was performed including beta-amyloid (DAKO M0872 1:200), tau (DAKO A0024 1:50,000), ubiquitine (DAKO Z0458 1:5,000), and alpha-synuclein (Zymed zs 18-0215 1:2,000). The adult subjects had no neurological or psychiatric disorders, and the neuropathological screening revealed no pathology such as bleeding, infection, degeneration, tumors, or metastases in any part of the cerebrum or cerebellum. There was no indication of Alzheimer's disease or other neurodegenerative disorders. Likewise, all newborn subjects were without malformations, known chromosomal abnormality, hydrops, sepsis, or any other disease involving the central nervous system; see Tables 1 and 2 for specimen characteristics. The autopsies on the newborn brains were usually performed within 1 or 2 days after death, but the time until fixation of the brains was not consistently recorded. The autopsies were sometimes delayed and thus performed up to 3.5 days after death. From death until autopsy, the subjects were refrigerated at 50° C in order to minimize cellular degeneration. Each newborn had a normal birth weight with a mean fetal growth index of 1.04 (0.93–1.20), which is observed birth weight divided by expected mean birth weight (Larsen et al. 1990); see Table 1. Such a ratio-based classification was preferred to more traditional forms based on percentiles or standard deviations (SDs) because it conveys important clinical and statistical information, giving the percentage of weight relative to the mean.

The brains were obtained from necropsies from 1975 to 1997 in accordance with the Danish laws on autopsied human tissue, and material was collected after parental consent and with approval of the local ethical committee.

Tissue Processing

The brains were fixed in 4% buffered formaldehyde for at least 5 months. Soft and vulnerable newborn brains were further fixed in saturated picric acid for 2–4 weeks to harden the tissue prior to cutting. The left or right hemisphere was selected systematically at random. The newborn hemispheres and underlying diencephalons were cut into 2–3 slabs (5–7 for adult brains), each 2.5 cm thick, before embedding in paraffin. The initial thick slicing was effectively random due to the central and therefore hidden location of the thalamus. The slabs were cut into 40-μm thick coronal sections on a sledge microtome, and a known and predetermined fraction of the sections was sampled serially throughout the hemisphere. A moistened filter paper was placed on the paraffin block to sample the section without using a water bath. When the section adhered to the filter paper, it was mounted on a double silane coated large glass slide. This was done by placing the paper on the slide and applying gentle pressure with a printing roller. This prevents the production of artifacts from tissue deformation, which can occur within the water bath, and consequently, the tissue section height postprocessing is mostly very close to 40 μm. Rarely, it may result in swelling of the sections (for illustration, see Eriksen and Pakkenberg 2007). The sampled sections were then instantly dried at 40 °C for 24 h, heated to 60 °C for 30 min prior to staining, dewaxed in xylene for 45 min, followed by 15 min in 99% ethanol, 10 min in 96% ethanol, 5 min in 70% ethanol, and 5 min in distilled water. The sections were stained using a modified Giemsa stain containing 25-mL Giemsa stock solution ("Merck," Darmstadt, Germany, product 1.09204) and 250-mL potassium hydrogen phosphate at pH 4.5, which was filtered before use. Finally, the sections were differentiated with 0.5% acetic acid and dehydrated through 96% ethanol for 1–5 min, 99% ethanol for 5–10 min, and xylene for 15 min.

Table 1
Demographic and clinical characteristics of newborn (gestational age, weeks + days) and adult subjects (age, years)

Subjects	Sex	Age	Body height (cm)	Body weight (kg)	Fetal growth ratio
Newborns					
1	Female	38 + 6		3.65	1.10
2	Male	40 + 5		3.93	1.03
3	Male	39 + 2		3.67	1.04
4	Male	41 + 3		3.86	0.97
5	Female	40 + 4		3.77	1.04
6	Male	38 + 0		2.90	0.93
7	Female	41 + 0		3.86	1.04
8	Male	41 + 0		4.66	1.20
Mean		40 + 1		3.79	1.04
Adults					
1	Male	43	175	82	
2	Female	64	154	50	
3	Male	65	173	85	
4	Female	70	169	54	
5	Male	71	175	78	
6	Female	75	159	60	
7	Female	75	172	56	
8	Female	85	159	39	
Mean		68.5	167	63	

Table 2
Postmortem data of newborn and adult subjects

Subjects	Cause of death	Brain weight (g)	Placental and maternal condition	Terminal period (days)	Postmortem interval (h)	Fixation period (months)	Hemisphere sampled
Newborns							
	Acute asphyxia	402	Normal			9	R
	Acute asphyxia	482	Tightening of true cord knot			45	R
	Acute asphyxia	442	Normal			10	R
	Isolated lung hypoplasia	441	Normal			54	L
	Acute asphyxia	407	Normal			18	L
	Pneumonia	144[a]	Normal			20	L
	Acute asphyxia	558	Normal			22	L
	Acute asphyxia	471	Chorioamnionitis			102	L
Mean		436				35	
Adults							
	AMI	1560			24	65	R
	AMI	1240		0, 1	24	90	R
	AMI	1420		0, 1	24	66	R
	AMI	1334		0, 1	33	96	R
	AMI	1380		0, 1	48	12	R
	AMI	1150			24		R
	Congestive heart failure	1138		0, 1	35	91	L
	AMI	1180		14	24	129	R
Mean		1300			29, 5	78	

Note: AMI, acute myocardial infarction; R, right; L, left.
[a]Weight of one hemisphere, not included in the mean.

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Delineation of MD

In order to estimate the total number of cells in the MD, a clear definition of the boundaries of this nucleus was necessary. As previously mentioned, both this definition and the definition of its subregions vary in existing literature (Dewulf 1971; Hirai and Jones 1989; Armstrong 1990; Jones 1997a). In this study, we chose anatomical criteria based on cell size, packing density, and relation to the internal medullary lamina applying the delineations made according to Hirai and Jones. Initially, the sections were studied through an Olympus Stereomicroscope VMZ 1X and 4X, and the MD and the subregions were demarcated directly on the coverslip using a fine permanent ink liner. The internal border of demarcations represented the specified area. Sections from 8 of the brains (6 newborn, 2 adult) were reevaluated by one of us (E.G.J.) and adjusted accordingly. For examples of the delineation of the subregions in the newborn MD, see Figure 1. An illustration of the delineation in the adult brain is given in Popken et al. (2000), which resembles the one in the newborns. The final delineations were made immediately prior to counting using the Computer Assisted Stereological Test Grid Software (CAST) Grid version 2.0 stereological software, Olympus, Denmark.

Definitions

The internal medullary lamina surrounds the anterior, lateral, ventral, and posterior surfaces of the large MD and also sends extensions over much of its dorsal and medial surfaces. The MD begins just caudal and ventral to the anteroventral nucleus and ends at the level of the rostral pulvinar. Its medial border is near the lateral wall of the third ventricle except where the paraventricular nucleus is interposed. The large cells of the CL clearly delineate the lateral boundary of the MD for most of its rostral–caudal extent. It is bounded posteriorly by the central medial nucleus. The lateral ventricle, fornix, anterior nuclei, and laterodorsal nucleus lie on its superior surface and are clearly recognizable. Most posteriorly, the inferior limit merges with the parafascicular nucleus and the center median nucleus. Given these relatively distinct features, delineating the external border of MD presented few difficulties, except perhaps at the most posterior level, adjacent to the medial part of the pulvinar, where the internal medullary lamina breaks up into clusters that invade the MD. Interior divisions of the MD include the MDmc, the MDpc, and the MDdc subregions. MDmc occupies the anteromedial aspect of MD and is composed of relatively large, deeply stained neurons in a densely packed homogeneous neuropil. Most of the dorsolateral aspect of MD is taken up of the MDpc, which possesses somewhat smaller neurons and a looser neuropil that is broken by bundles of fibers (Dewulf 1971; Hirai and Jones 1989). MDdc, in which we include the paralamellar part of the MD, envelops MD laterally and posteriorly (Jones 1997a). Its neurons are densely stained and resemble those of the CL.

Cell Types

We chose to identify large neurons and small neurons (presumably projection and local circuit neurons, respectively) and glial cells morphologically although the distinction between the largest glial cells and the smallest neurons is not a trivial problem. In this study, the neurons could be distinguished from astrocytes and oligodendrocytes in Giemsa-stained sections using a combination of criteria such as diffuse and even chromatin pattern, size and shape of the nucleus, a clearly visible nucleolus, and surrounding cytoplasm. The major morphological difference between the γ aminobutyric acidergic (GABAergic) interneurons and the excitatory relay cells is their size (Dewulf 1971; Armstrong 1990; Jones 1997c). The analysis of the thalamic neurons by size shows a bimodal frequency distribution (Dewulf 1971; Dorph-Petersen et al. 2004) of which the larger neurons are thought to be predominantly relay neurons. About 40% of the neurons in MD are small and represent predominantly interneurons (Dewulf 1971; Dorph-Petersen et al. 2004). In practice, it was possible to distinguish between large neurons being intensely stained, with large blue nuclei and matching large somal sizes, and small pale blue neurons with less cytoplasm and thereby differentiate between subtypes of neurons, see Figure 2 for an example. Glial cells, that is astrocytes, oligodendrocytes, and microglia, were counted as one entity.

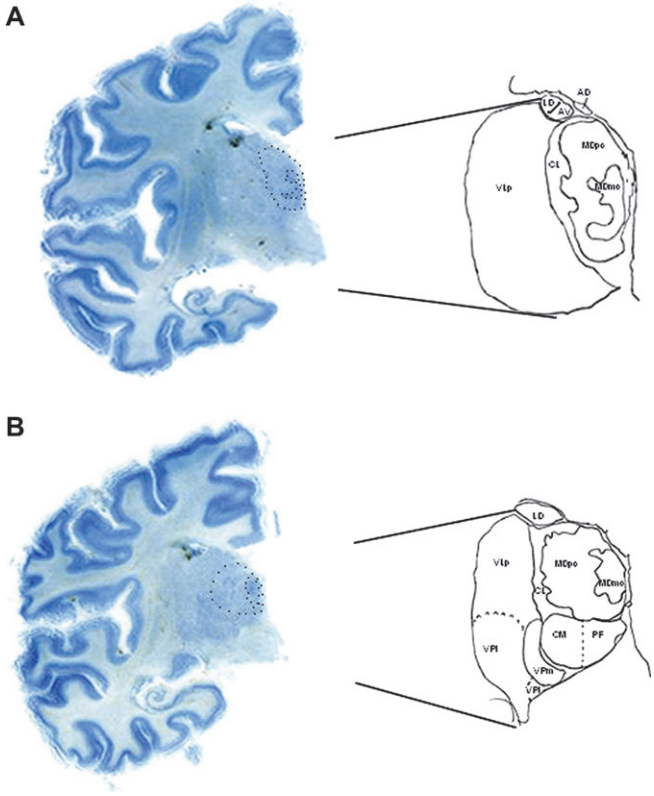


Figure 1. Photomicrographs showing the delineation of the subregions in the newborn MD at 2 different coronal positions, 3 mm apart, (*A*) anterior and (*B*) posterior. AV, anteroventral nucleus; AD, anterodorsal nucleus; LD, laterodorsal nucleus; CM, center median nucleus: PF, parafascicular nucleus; VLp, ventral lateral posterior nucleus; VPL, ventral posterior lateral nucleus; VPM, ventral posterior medial nucleus; VPI, ventral posterior inferior nucleus.

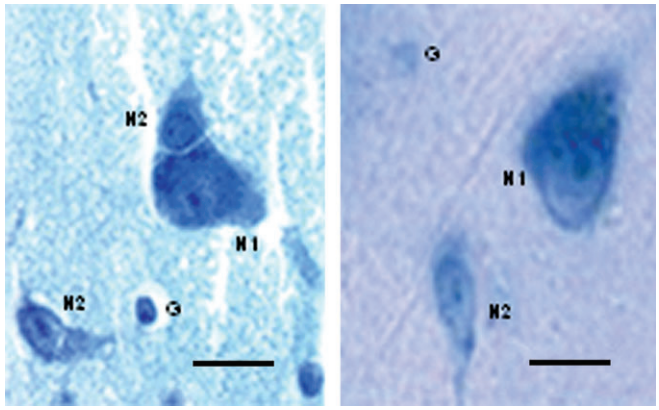


Figure 2. Photomicrographs of Giemsa-stained sections showing examples of large (N1) and small (N2) neurons and glial cells (G) in the MD thalamus. Left panel is from a newborn brain and right panel is from a 65-year-old male subject. Scale bar = 10 μm.

Counting Method

Stereology is a set of methods for accessing information regarding the number, length, surface area, and volume of specimen attributes based on mathematically unbiased principles (Gundersen 1986). It is ideal for this particular study as it provides a system of quantifying 3-dimensional (3D) microscopic structures, that is neurons and glial cells, through analysis of a 3D probe in a thick histological section. The method is in principle unbiased due to there being no underlying assumption made about the structure being studied. The method relies on the systematic uniform random sampling (SURS) of the specimen, yielding an estimate

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of the characteristic being sampled obtained with an accuracy determined by the stringency of the investigators.

The design applied to this study, the optical fractionator (West et al. 1991), essentially combines the stereological principles of the optical disector (Sterio 1984) and the fractionator sampling scheme (Gundersen 1986; Pakkenberg and Gundersen 1988). The disector is basically a 3D probe, of an exact predefined volume, that enables us to sample objects with a probability that is proportional to their number and not their size (West 1993). The fractionator principle provides a method by which a quantification of the total cell number of a structure can be obtained. This is unaffected by even considerable tissue changes such as shrinkage during fixation of the human newborn brain. It is a requirement that the entire undamaged structure, in this case the thalamus, is available for sampling (Gundersen 1986).

For this study, either the left or the right MD was chosen, and a known section sampling fraction (ssf) of all sections was sampled systematically with a random start. In the newborn brains, 1/20–1/25th of the sections were sampled whereas in the adult's every 1/40–1/60th section. This provided an average of 11 sections per MD, which in turn provided around 8 (range 3–13) sections for further analysis in each subregion.

The counting equipment consisted of an Olympus BX50 microscope with a 100× oil-immersion objective with a high numerical aperture (1.40) connected to a Heidenhain microcator and a motorized specimen stage PRIOR proscan. A video camera and a computer running CAST Grid version 2.0 were also used. The cells were counted directly in a known fraction of each region using the optical disector in a systematically random pattern. The area of the 2-dimensional counting frame was verified relative to the area associated with each movement in the *x*-*y* direction, the area sampling fraction (asf). The size of the counting frame and the *x*-*y* distance were adjusted in each brain to count approximately 100–200 cells of each type per region. Consequently the *x*-*y* interval varied from 900–1000 μm in MDpc to 500–600 μm in MDmc and MDdc. The fixed height of the disector, *h*, was chosen to be 20 μm, and the mean section thickness, *t*, in each brain was estimated from measurements of *t* made at every fourth disector containing sampled cells. The section thickness was determined with the 100× oil-immersion objective by focusing from the upper to the lower surface of the section at the sampled disectors. The sections were cut at 40 μm although shrinking during subsequent processing (Dorph-Petersen et al. 2001) providing a final mean section thickness of 38 μm (range 31–41 μm). To exclude the possibility of lost cells, and because all cells in focus at the upper plane of the disector were not included in the counts, an upper and lower guard zone were introduced of approximately 5 μm and 15 μm, respectively. To ensure that the problem with lost caps could be excluded, cells in the full height of the section were recorded in a pilot study and showed a stable 3D density within the height of the disector. No difference was found in the average section thickness between groups. The nucleolus was used as the counting item for neurons, whereas the nucleus was used for glial cells.

A "Heidenhain" microcator was used to measure movements in the *z* axis with a precision of 0.5 μm. The height sampling fraction (hsf) was calculated from

$$\text{hsf} = \frac{h}{t_{Q^-}} \text{ for } t_{Q^-} = \frac{\sum_i (t_i q_i^-)}{\sum_i q_i^-},$$

where t_i is the local section thickness centrally in the *i*th counting frame with a disector cell count of q_i (Dorph-Petersen et al. 2001).

For one nucleus of the thalamus, the sum of cells counted in all the disectors was denoted $\sum Q^-$. The total number of cells, *N*, in the particular subregion of MD was calculated from

$$N = \frac{1}{\text{ssf}} \times \frac{1}{\text{asf}} \times \frac{1}{\text{hsf}} \times \sum Q^-.$$

The same investigator carried out all counting in the newborn brain, whereas a second investigator, after having established an interrater repeatability of 94%, performed some of the counting of the adult MD.

Precision of the Estimate

The precision of the estimated means, that is the intraindividual stereological error, was predicted as the coefficient of error (CE), which is caused by sample error related to SURS and counting noise (Gundersen et al. 1999). In order to minimize the workload associated with the counting process and still achieve an acceptable CE, a pilot study was used to determine the optimal *x*-, *y*-step lengths for each subregion of the MD. A mean of 538 neurons were counted in an average of 539 disectors per MD. This was a result of 258 neurons in an average of 269 disectors in MDpc, 117 neurons in an average of 123 disectors in MDmc, and 163 neurons in an average of 147 disectors in MDdc. The CE indicates the precision of the estimate of a single region. The total CE was estimated from the formula below combined with a contribution from the CE of the section thickness:

$$\text{CE}(N) = \frac{\sqrt{\text{Var}_{\text{SURS}} + \text{Noise}}}{\sum Q^-}.$$

Noise is the sum of counted particles and Var_{SURS} is the estimator variance under SURS. The Var_{SURS} (*N*) is obtained from the formula

$$\text{Var}_{\text{SURS}} \text{ for } N = \frac{(3(A - \text{Noise}) - 4B + C)}{240}.$$

The systematic section series of particles count are denoted $f_1, f_2, \ldots, f_n$ and $A = \sum_{i=1}^{n} f_i^2$, $B = \sum_{i=1}^{n-1} f_i f_{i+1}$, and $C = \sum_{i=1}^{n-2} f_i f_{i+2}$ (for further details, see equations [20–22], Gundersen et al. 1999).

Statistical Analysis

The results of the 2 groups were compared with regard to their mean using the unpaired 2-tailed Student's *t*-test with a 2*P* significance level of 0.05.

Results

Estimates of the total neuron number for the entire MD and its subnuclei from the 8 infant and 8 adult brains are shown in Table 3 and Figure 3. In the adults, the total number of neurons in the entire MD was an average of 41% lower than in the newborn, a difference that was statistically highly significant ($P < 0.001$). The estimated average total number of neurons in MD of the newborns was 11.2 million (coefficient of variation [CV] = SD/mean = 0.16, SD = 1.79), compared with the adults 6.43 million (CV = 0.15, SD = 0.96).

The difference in total neuron number was not homogeneous across MD. In the adult brains, MDpc and MDdc demonstrated a significantly lower neuron number compared with the newborns by 53% ($P < 0.001$) and 33% ($P = 0.024$), respectively. In MDmc, no significant difference was found ($P = 0.43$).

Comparing the number of large, presumed projection neurons and small, presumed local circuit neurons between the 2 groups also provided a statistically significantly difference ($P \leqslant 0.001$). We find that 37% of the neurons in the adult MD are small, presumed GABAergic interneurons whereas 51% are small in the newborn MD.

Table 3
Results of the stereological analyses

	Entire MD N_{total} (10^6)	Entire MD N1 (10^6)	Entire MD N2 (10^6)	MDpc N_{total} (10^6)	MDmc N_{total} (10^6)	MDdc N_{total} (10^6)	Entire MD Glial cells (10^6)
Newborns							
Mean	11.2	5.65	5.58	8.48	1.35	1.41	10.57
CV	0.16	0.15	0.23	0.22	0.29	0.30	0.15
Adults							
Mean	6.43	4.02	2.41	3.95	1.54	0.94	36.33
CV	0.15	0.16	0.19	0.26	0.29	0.28	0.36
P value	<0.001	<0.001	<0.001	<0.001	0.43	0.024	<0.001

Note: N, neurons; N1, large neurons; N2, small neurons; G, glial cells, CV = SD/mean.

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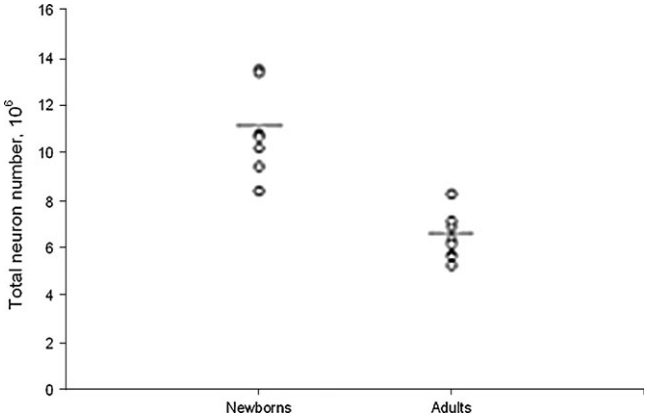


Figure 3. Scatter plot showing the estimates of the total number of neurons in the entire MD thalamus for the newborn and adult subjects.

Glial cell numbers were as expected much higher in the adult brains with an increase of almost 4 times from 10.6 million (CV = 0.15) at birth to 36.3 million (CV = 0.36) in adult life ($P \leqslant 0.001$).

Discussion

We found that the MD had significantly more neurons and fewer glial cells at the time of birth compared with the adult brain. The increase in glial cell number came as no surprise because other brain regions including the neocortex show the same pattern (Larsen et al. 2006), while the higher neuron number was unprecedented.

Comparing with other stereological studies of the MD, our mean unilateral total neuron number lies higher for the adult controls than in all but one (Dorph-Petersen et al. 2004) of the previous studies. As noted by Dorph-Petersen et al. (2004), our estimate of the total number of large neurons closely resembles that of the total number of neurons found earlier. Difficulties in differentiating large glial cells from the small neurons on the basis of different staining procedures and optical properties may account for discrepancy among different groups. Furthermore, we cannot exclude the possibility that difficulty in identification of small neurons versus glial cells increases with the age of the subjects, but the nerve and glial cells look remarkably the same in the newborn brain compared with the adult, so this influence is probably very small. Other limitations include that the left hemisphere, due to technical quality, was overrepresented in the newborns compared with the adult brains. However, Pakkenberg (1992) showed no difference in the volume of MD comparing left to right, which may strengthen although not prove the theory that the reduced cell numbers in the adult brains are not due to lateralization.

We have previously estimated the total cell numbers of MD obtaining much lower numbers (Pakkenberg 1990). In comparing the 2 studies, obvious differences in method and design must be taken into account. The delineations made in our first study were different from the present because myelin staining in neighboring sections was used to indicate the borders of the MD. This made the entire volume of MD smaller. Given that both the total neuron number and the volume of the MD lie considerably lower in our first study than any other study, this delineation technique substantially accounts for that difference. Secondly, the first study applied the physical disector, counting cells in 4-μm thick sections at a final magnification of 443×, where the image was projected onto a table, a counting setting that was less optimal than the present and again explaining that the small neurons were possibly not included in the counts.

It is not known whether the postnatal loss of almost half of the neurons in the entire MD is primary or secondary to structural changes in the prefrontal cortex or regions afferent to the nucleus. In the adult brain, neuronal loss in the thalamus occurring as a result of cortical pathology is commonly accompanied by gliosis, whereas transneuronal degeneration or deafferentation is accompanied by shrinkage of cells, 2 phenomenons not encountered in this study (Lowe et al. 1997; Edmonds et al. 1999). In contrast, neurons lost during brain development are less likely to be accompanied by any recognizable changes in surviving neurons or glial cells (Jones 1997b). It could result from failure of neurons in the MD to establish synaptic connections in the cortex or just from a selected degree of normally programmed cell death in the thalamus.

Studies of gray matter maturation have recorded loss of cortical gray matter over time (Sowell et al. 2001, 2004; Gogtay et al. 2004), which is discussed to be due to myelination and increased synaptic pruning during adolescence and early adulthood (Bourgeois et al. 1994). Accordingly, magnetic resonance imaging scans have shown regression of subcortical gray matter, including the thalamus, during development from childhood through adolescence (Caviness et al. 1996; Szaflarski et al. 2006).

An overproduction and elimination of neurons have been found in the primate thalamus. Williams and Rakic (1988) studied the timing, magnitude, and spatial distribution of neuron elimination in the dorsal lateral geniculate nucleus of 57 rhesus monkeys (Macaca mulatta) ranging in age from the 48th day of gestation to maturity. They found a large number of geniculate nuclei neurons to be eliminated over a 40- to 50-day period spanning the middle third of gestation. Very few neurons were lost after embryonic day 100, and as early as embryonic day 103, the number had fallen to the adult average. It was concluded that the number of geniculate neurons might play a key role in controlling the number of the retinal cells that survive to maturity.

Understanding the neural basis of cognition and establishing a causal link between biological structure and cognitive and affective function have been of interest for decades. As described by Changeux and Dehaene (1989), the mechanism of stabilization (and elimination) of synaptic connections by spontaneous and/or evoked activity in developing neuronal networks is decisive. This contributes to the shaping of the adult connectivity within an envelope of genetically encoded forms. The selection process by which some neurons survive and other neurons undergo apoptosis during development is thus complex. More recently, neurotrophins, such as nerve growth factor, have been recognized to strongly affect the development and plasticity of the visual system in both thalamus (lateral geniculate nucleus) and cortex (Wahle et al. 2003), but whether the same mechanism contributes to effects in MD remains to be elucidated. However, it seems evident that the process must be controlled strictly for the MD to function normally because an individual with an increased total number of cells might suffer from depression, whereas subjects with an abnormal reduction may experience psychosis (Pakkenberg 1990, 1992; Popken et al. 2000; Young et al. 2000, 2004; Byne et al. 2002). Studies of normal human brain development might help understand complex brain pathology as seen in schizophrenia and mood disorders.

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Notes

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